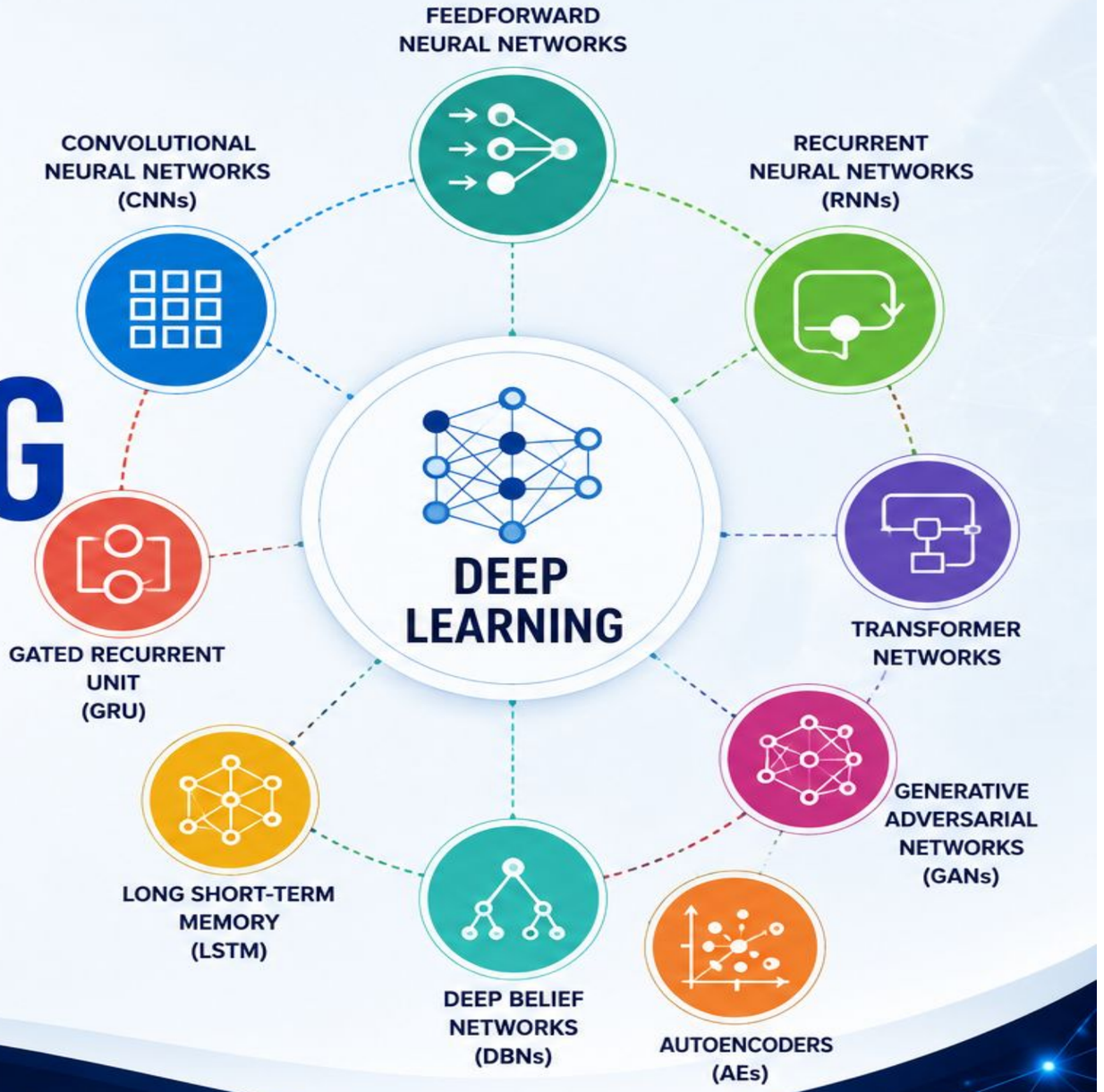
TYPES OF DEEP LEARNING

Exploring the Diverse Architectures Powering Intelligent Systems





Topics :

- Types of Deep Learning

INTRODUCTION

- **Deep learning networks** are **artificial neural network architectures** capable of **learning complex patterns** from **large amounts of data**.
- **Different architectures** are designed for **different types of data** such as **images, text, speech, videos, graphs, and time-series data.**



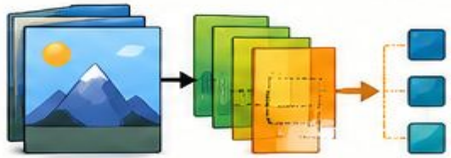
TYPES OF DEEP LEARNING MODELS

01 Feedforward Neural Network (FNN)



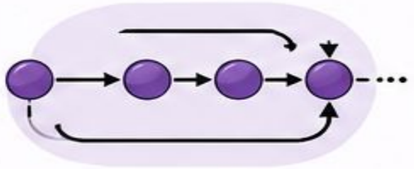
Information flows in one direction from input to output without loops. Used for basic classification and regression tasks.

02 Convolutional Neural Network (CNN)



Automatically extracts spatial features using convolution operations. Best suited for image and video analysis.

03 Recurrent Neural Network (RNN)



Processes sequential data by maintaining a hidden state or memory using loops. Useful for text, speech and time series.

04 Long Short-Term Memory (LSTM)



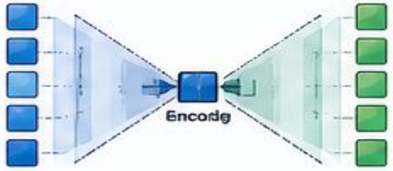
Special type of RNN that captures long-term dependencies and remembers important information for longer periods of time.

05 Gated Recurrent Unit (GRU)



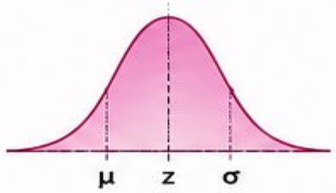
A simplified version of LSTM with fewer gates. Controls information flow efficiently with less computational cost.

06 Autoencoder (AE)



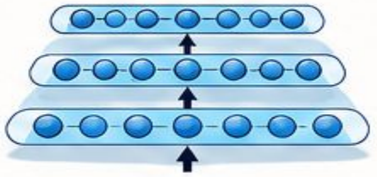
Learns a compressed representation of input data and then reconstructs the original data. Used for dimensionality reduction and feature learning.

07 Variational Autoencoder (VAE)



Probabilistic autoencoder that learns data distribution in the latent space and can generate new, similar data samples.

08 Deep Belief Network (DBN)



Composed of multiple layers of Restricted Boltzmann Machines (RBMs). Learns hierarchical feature representations.

09 Generative Adversarial Network (GAN)



Generator (creates fake data)

Discriminator (distinguishes real/fake)

Two networks compete with each other. Generator creates realistic data, while Discriminator distinguishes real from fake.

10 Transformer



Uses self-attention mechanism to process sequential data in parallel. Widely used in NLP, translation, and many other applications.

Deep learning models are designed to solve different types of problems such as **classification, prediction, generation,** and **understanding patterns** in data like images, text, audio, video, and time series.

1. Feed Forward Neural Networks(FNNs)

FNNs, also known as Multilayer Perceptron's (MLPs), are the **simplest type of artificial neural network.**

- In an FNN, **information moves in only one Direction**. It can be **moved from the input layer to the output layer**. There are **no back-loops in the feed-forward network**.
- To **minimize the prediction error**, the **back propagation algorithm can be used to update the weight values.**
- FNNs are widely used for **simple classification and regression problems.**

Applications:

- Data Compression
- Pattern Recognition
- Computer Vision
- Sonar Target Recognition
- Speech Recognition
- Handwritten Characters Recognition

2. Convolutional Neural Networks(CNNs):

- **CNNs** are a type of **deep learning model** that are especially effective for **processing grid-like data** such as **images, making them extremely useful for image recognition, image classification, clustering of images, and object recognition etc.**
- **CNN is a specialized neural network** designed to **process image data**. It **automatically extracts spatial features** such as **edges, textures, shapes, and objects.**
- An advanced example of CNN use is in **self-driving cars**, specifically in the system **responsible for detecting objects around the vehicle.**
- The **real-time application** where the **CNN doesn't just classify static images** but **continuously processes video frames** to **identify and classify** objects **such as other vehicles, pedestrians, traffic signs, and lanes.**

01



Identify Faces, Street Signs, Tumors

Detect and locate objects such as faces, street signs, and tumors in images.

02



Image Recognition

Recognize and classify objects, people, places, or patterns in images accurately.

03



Video Analysis

Analyze video content for activity recognition, tracking, and behavior understanding.

04



NLP (Natural Language Processing)

Understand, interpret, and generate human language (text or speech).

05



Anomaly Detection

Identify unusual patterns or outliers in data for fraud, faults, or security threats.

06



Drug Discovery

Accelerate the discovery of new drugs by predicting molecular interactions and compound effectiveness.

07



Checkers Game

Deep learning agents can learn strategies and play games like checkers, chess, etc.

08



Time Series Forecasting

Predict future values based on historical data in areas like finance, weather, and sales.

3. Recurrent Neural Networks(RNNs)

- RNNs are designed for sequential data where previous information influences future predictions.
- **Recurrent neural networks** get **feedback from output neurons.** RNNs are designed to **recognize patterns in sequences of data, such as text, genomes, handwriting, or the spoken word,** and are therefore **very effective for natural language processing and speech recognition tasks**.
- Before Transformers came to age, **a special variant of RNN Long Short-Term Memory Networks (LSTM)** was prevalent in **machine translation and text generation**.

- The Recurrent neural network mainly accesses the **preceding info of existing iterations**.
- **For example**, to **guess the succeeding word in any sentence**, **one must have knowledge about the words that were previously used**. It **not only processes the inputs** but **also shares the length as well as weights crossways time**.

Applications:

- Machine Translation
- Robot Control
- Time Series Prediction
- Speech Recognition
- Speech Synthesis
- Time Series Anomaly Detection
- Rhythm Learning
- Music Composition

4. Long Short-Term Memory (LSTM)

- **Long Short-Term Memory (LSTM)** is a special type of **Recurrent Neural Network (RNN)** **designed to learn** and **remember information over long periods.**
- It overcomes the **vanishing gradient problem** of **traditional RNNs** by using **memory cells** and **gating mechanisms** to selectively **retain important information and discard irrelevant information.**
- By maintaining a memory cell and regulating information flow through the **Forget Gate**, **Input Gate**, and **Output Gate**, LSTM **preserves useful information for a longer duration and improves learning efficiency**.

Applications:

- Machine Translation
- Text Prediction
- Sentiment Analysis
- Speech Recognition
- Medical Diagnosis
- Time Series Anomaly Detection
- Music Composition

LSTM Applications



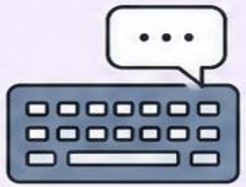
Speech Recognition

Converts spoken language into text.



Machine Translation

Translates text from one language to another.



Text Prediction

Predicts the next word while typing.



Sentiment Analysis

Determines whether text is positive, negative, or neutral.



Time-Series Forecasting

Predicts future values such as stock prices, weather, or sales.



Chatbots & Virtual Assistants

Understands user conversations and generates responses.



Medical Diagnosis

Analyzes sequential medical data to detect diseases.



Video Captioning

Generates text descriptions for videos.



Music Generation

Learns musical patterns and composes new music.

5. Gated Recurrent Unit (GRU)

- **Gated Recurrent Unit (GRU)** is a special type of **Recurrent Neural Network (RNN)** introduced **to simplify the architecture** of **Long Short-Term Memory (LSTM)** networks.
- It is designed to **learn long-term dependencies** in **sequential data** while using **fewer gates** and **fewer parameters**, making it faster and more computationally efficient than LSTM.
- Unlike **LSTM, which uses three gates** (Forget Gate, Input Gate, and Output Gate) and a separate memory cell, **GRU uses only two gates**—the **Update Gate** and the **Reset Gate**.
- These **gates control the flow of information** and **enable the network** to decide what **information to retain and what information to forget**.

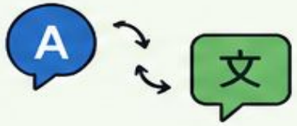
GATED RECURRENT NETWORKS (GRN) APPLICATIONS

1. Speech Recognition



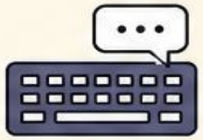
Converts spoken language into text.

2. Machine Translation



Translates text from one language to another.

3. Text Prediction



Predicts the next word while typing.

4. Sentiment Analysis



Determines whether text is positive, negative, or neutral.

5. Time-Series Forecasting



Predicts future values such as stock prices, weather, or sales.

6. Chatbots & Virtual Assistants



Enables intelligent conversational systems.

7. Handwriting Recognition



Converts handwritten text into digital text.

8. Medical Diagnosis



Analyzes sequential medical data to detect diseases.

9. Anomaly Detection



Detects unusual patterns in data and signals.

10. Financial Market Prediction
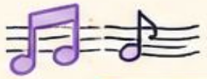


Forecasts stock prices, exchange rates, and market trends.

11. Video Captioning



Generates descriptive captions for videos.

12. Music Generation



Learns musical patterns and composes new music.

13. Machine Learning in IoT



Processes sequential sensor data for smart IoT applications.

14. Natural Language Processing (NLP)



Used in text summarization, question answering, and language modeling.

6. AUTOENCODERS

- **An autoencoder** is an **unsupervised neural network** used to **compress input data into a lower-dimensional representation** and **reconstruct it.**
- They are **composed of two parts**: **An encoder**, which **transforms the input data into a lower-dimensional code**, and a **Decoder,** which attempts to **reconstruct the original input data from this code.**
- Auto encoders are often employed for **anomaly detection in various types of data,** from **time-series sensor data to complex images**

Applications:

- Classification.
- Clustering.
- Feature Compression.

7. VARIATIONAL AUTOENCODERS(VAE)

- A **Variational Autoencoder (VAE)** is a and an advanced version of the **Autoencoder (AE)**. **probabilistic deep learning model**
- Unlike a **traditional Autoencoder** that learns **a fixed compressed representation of the input,** a VAE learns a **probability distribution (mean and variance) of the latent space**.
- This enables it to generate **new and realistic data samples** that are **similar to the training data.**

A VAE consists of two main parts:

- **Encoder:** Converts the **input data into a probability distribution** (latent space).
- **Decoder: Generates new data samples by decoding a randomly sampled latent vector** from this distribution.
- Because of its ability to generate new data, VAE is widely used in **image generation, face synthesis, anomaly detection, medical image analysis, and drug discovery**

8. Deep Belief Networks(DBN)

- A **Deep Belief Network (DBN)** is a type of **deep neural network** that consists of **multiple hidden layers arranged one above another.**
- **Each layer learns increasingly complex features** from the **output of the previous layer**, **enabling the network** to extract **hierarchical representations of the input data**.
- The **lower layers learn simple patterns**, while the **higher layers learn more abstract and meaningful features.**
- DBNs **first learn useful features from the input data** using **unsupervised learning**. After learning these features, the **network is fine-tuned using supervised learning** to **improve its performance in classification or prediction tasks.**
- This **combination of feature learning and fine-tuning allows** DBNs to model **complex data effectively.**

Applications

1. **Image Classification** – Classifies images into different categories (e.g., cats vs. dogs).
2. **Speech Recognition** – Converts spoken language into text.
3. **Medical Image Analysis** – Detects diseases from X-rays, CT scans, and MRI images.
4. **Anomaly Detection** – Identifies unusual or abnormal patterns in data, such as fraud or equipment faults
5. **Recommendation Systems** – Recommends products, movies, or music based on user preferences.
6. **Natural Language Processing (NLP)** – Used for text classification, document categorization, and sentiment analysis.

9. Generative Adversarial Networks(GAN)

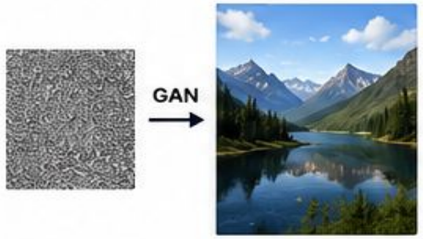
- A **Generative Adversarial Network (GAN)** is a type of **deep learning model** used to **generate new and realistic data samples** that are **similar to the data used during training**.
- GANs consist of **two neural networks that compete with each other**:

  1. Generator (G): Creates new data samples (such as images, videos, or audio).

  2. Discriminator (D): Evaluates whether the generated data is real (from the training dataset) **or fake** (generated by the Generator).
- **During training**, the **Generator continuously improves its ability to create realistic data**, while the **Discriminator becomes better at distinguishing between real and generated data**. This competition helps the **Generator produce highly realistic outputs.**

GAN APPLICATIONS

1. Image Generation



Generates realistic images of objects, scenes, and natural environments.

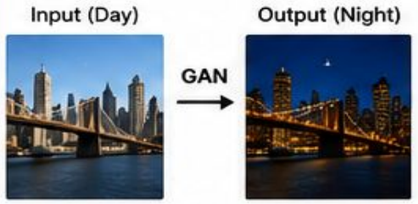
2. Face Generation



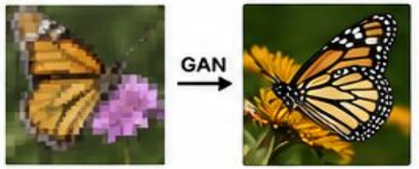
Creates realistic human faces that do not exist in real life.

3. Image-to-Image Translation

Convert images from one domain to another.



4. Super-Resolution Imaging



Enhances the resolution and quality of low-resolution images.

5. Image Inpainting (Restoring Missing Parts)



Fills in missing, damaged, or occluded parts of an image realistically.

6. Data Augmentation



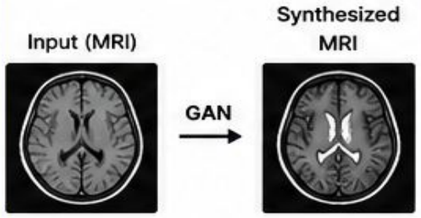
Generates additional synthetic data to increase the size and diversity of training datasets.

7. Video Generation



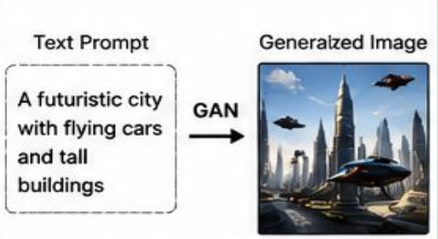
Generates realistic and coherent video sequences frame by frame.

8. Medical Image Synthesis



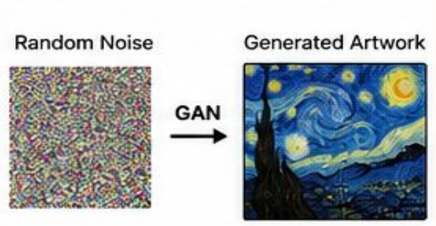
Generates synthetic medical images for training and research purposes.

9. Text-to-Image Generation



Creates images from textual descriptions and natural language prompts.

10. Artwork and Design Generation



Generates paintings, art, logos, sketches, and designs in creative fields.

Key Point: GANs learn the underlying data distribution and generate new, realistic data similar to the training dataset.

10. TRANSFORMERS

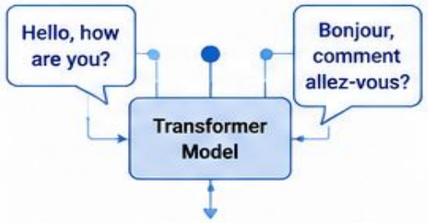
- Transformers, specifically the **"self-attention"** mechanism within them, have been a **game-changer in the field of deep learning.** They are especially powerful in handling **sequence data for tasks such as language understanding, translation, and text summarization**.
- The best-known implementation of a transformer is **OpenAI's GPT-3 model**, which can generate **human-like text given some input.** It can be used for tasks like **writing essays, answering questions, creating poetry, and even writing software code.**

Example

ChatGPT generates responses using Transformer architecture.

TRANSFORMER APPLICATIONS IN REAL TIME

1. Machine Translation



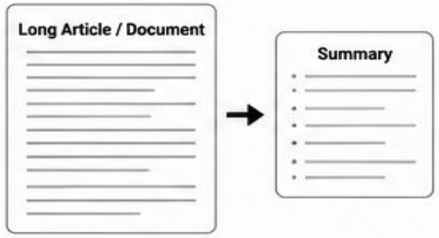
Transforms text from one language to another in real time.

2. Real-time Speech Recognition



Converts spoken words into text instantly.

3. Text Summarization



Summarizes long documents or articles in real time.

4. Real-time Chatbots & Virtual Assistants



Understands user queries and generates human-like responses instantly.

5. Real-time Text Generation (Auto-complete)



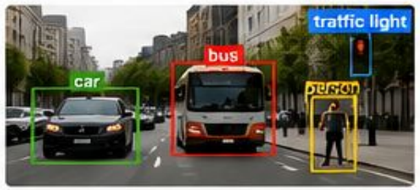
Predicts and suggests next words or sentences as you type.

6. Image Captioning



Generates descriptive captions for images in real time.

7. Real-time Object Detection (With Vision Transformer)



Detects and classifies objects in images or video streams instantly.

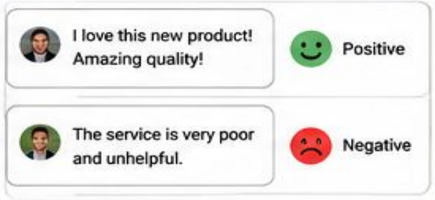
8. Video Understanding & Activity Recognition



Activity: Playing Tennis

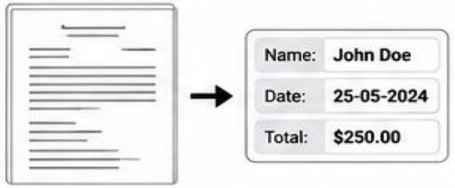
Recognizes actions or activities in videos in real time.

9. Real-time Sentiment Analysis



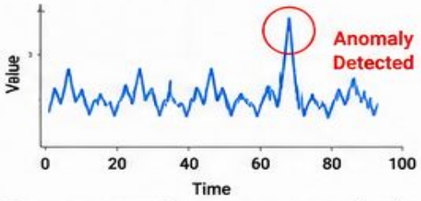
Analyzes sentiment of tweets, reviews, or comments instantly.

10. Document Information Extraction



Extracts key information from invoices, forms, or receipts in real time.

11. Real-time Anomaly Detection (With Time Series Transformer)



Detects unusual patterns or anomalies in sensor or network data instantly.

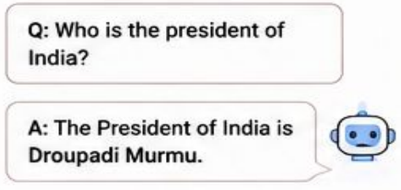
12. Code Generation (Auto-complete)

```python
def add(a, b):
    return a + b

def multiply(a, b):
    return * b
```

Suggests and generates code snippets in real time.

13. Real-time Question Answering



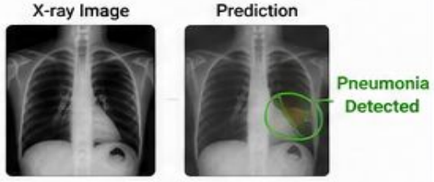
Answers factual questions instantly from large knowledge sources.

14. Real-time Recommendation Systems



Provides personalized recommendations for products, movies, or content.

15. Medical Image Analysis (With Vision Transformer)



Analyzes medical images (X-rays, MRI, CT scans) and gives results in real time.

Key Point: Transformers enable real-time understanding, generation, and decision-making across text, speech, vision, and time-series data.



Thank You!

Thank you for your attention
and for being a part of this
Deep Learning journey.

Deep Learning • Endless Possibilities

LEARN

UNDERSTAND

ANALYZE

INNOVATE

MODEL

TRAIN

PREDICT

IMPACT